

Integrable Probability and KPZ Universality

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1 Introduction & Overview

These notes were written for a reading course on integrable probability following a 2012 set of lecture notes on the subject by Alexei Borodin and Vadim Gorin [BG15]. Integrable probability is a relatively new field (emerging only in the last few decades) concerned with the application of algebraic and combinatorial techniques to probabilistic models. Its development has been largely motivated by the analysis of random growth models in the Kardar-Parisi-Zhang (KPZ) universality class. The focus of these notes is not to provide survey of the KPZ class and some of the known integrable models therein (for this, see [Zyg22]), but rather an overview of the some of the mechanisms by which the solutions to many models in the class are derived.

We begin with a discussion of point processes (and, in particular, determinantal point processes), which have utility in the asymptotic analysis of some growth models in the KPZ class. We then present the RS correspondence and its connection to the so-called Plancherel measure on the set of Young diagrams, which are intimately related to the polynuclear growth and last passage percolation models (among others) of the KPZ class. Finally, we then utilize the methods developed to present an asymptotic analysis of the Plancherel measure, leading to an instance of limit shape phenomena.

2 Point Processes

2.1 Definitions & Examples

Consider a “nice” space X – say, one which is separable and completely metrizable. A *point process* on X can be thought of as a random “sparse” collection of points in X ; “sparse” meaning that only finitely many points should be visible upon zooming in at any particular location (i.e. points do not accumulate anywhere).

Definition 2.1. A subset $X \subset X$ is called a *point configuration* in X if it is *locally finite* (i.e. has no accumulation points). The set of all point configurations in X shall be denoted by $\text{Conf}(X)$.

By definition, a point configuration in X has finite intersection with each compact $A \subset X$. To each such A , we can subsequently associate an integer-valued function $N_A: \text{Conf}(X) \rightarrow \mathbb{Z}_{\geq 0}$ which counts the number of points each configuration has in A , i.e. $N_A: X \mapsto |X \cap A|$. We give $\text{Conf}(X)$ the structure of a measurable space by endowing it with the smallest σ -algebra with respect to which each of the functions N_A is Borel-measurable, i.e. the σ -algebra generated by all $N_A^{-1}(k)$ where k ranges over the non-negative integers and A the compact subsets of X .

Definition 2.2. A *point process* on X is a probability measure on $\text{Conf}(X)$.

In alignment with our intuition about point processes as being random collections of points, it is common practice to refer to a point process $\mathbb{P}$ on X as just X , where X is understood to denote a random element of $\text{Conf}(X)$ distributed according to $\mathbb{P}$. Our first result about point processes is that they are determined uniquely by the finite-dimensional distributions of the counting functions:

Proposition 2.1. If point processes $\mathbb{P}$ and $\mathbb{Q}$ on the same space X are such that the N_A have the same marginal distributions under $\mathbb{P}$ as they do under $\mathbb{Q}$, then $\mathbb{P} = \mathbb{Q}$.

Proof. The assumption implies that the λ -system $\{\mathbb{P} = \mathbb{Q}\}$ contains the π -system

$$\Pi = \{(N_{A_1}, \dots, N_{A_n})^{-1}(k_1, \dots, k_n) : n \in \mathbb{N}, k_i \in \mathbb{Z}_{\geq 0}, A_i \subset X \text{ compact}\}.$$

By Dynkin’s theorem, it follows that $\{\mathbb{P} = \mathbb{Q}\} \supset \sigma(\Pi) \supset \sigma(N_A^{-1}(k) : k \in \mathbb{Z}_{\geq 0}, A \subset X \text{ compact})$. $\square$

The key takeaways from the above proposition are twofold. The first is that in order to describe a point process, it is sufficient to describe the finite-dimensional distributions of the N_A . We’ll see soon that this leads to more intuitive descriptions of specific examples of point processes. The second is that any given point process X can be identified with the family of random variables $\{N_A\}$. It is not hard to see that this family satisfies the following properties almost surely:

1. $N_\emptyset = 0$,
2. $N_A \leq N_B$ whenever $A \subset B$,

3. $N_{\cup_n A_n} = \sum_n N_{A_n}$ whenever $A_1, A_2, \dots$ are disjoint.

In other words, when viewed as a function of A , N_A looks like a random counting measure on $\mathcal{X}$. From this, we derive a second definition of a point process on $\mathcal{X}$ that is often useful when endeavouring to *construct* a specific process:

Definition 2.3. A *point process* on $\mathcal{X}$ is a random $\mathbb{Z}_{\geq 0}$ -valued Radon measure on $\mathcal{X}$ that almost surely assigns value at most 1 to singletons.

With the preliminary definitions out of the way, let's see some examples of point processes. The two most important are the following:

Example 2.1 (Bernoulli Process). Consider the case when $\mathcal{X}$ is discrete, say $\mathcal{X} = \mathbb{Z}^d$. Given $p \in (0, 1)$, the *Bernoulli process with parameter p* is the point process on $\mathcal{X}$ in which a point appears at each node $\mathbf{n}$ with probability p , independently of the other nodes. From a formulaic perspective, the Bernoulli process is the point process X defined by

$$X = \sum_{\mathbf{n} \in \mathbb{Z}^d} \delta_{\mathbf{n}} \mathbb{I}(X_{\mathbf{n}} = 1)$$

where $(X_{\mathbf{n}} : \mathbf{n} \in \mathbb{Z}^d)$ is a collection of i.i.d. Bernoulli p random variables.

Example 2.2 (Poisson Process). In this example we present the continuous analogue of the Bernoulli process. A *Poisson process* is a point process on $\mathcal{X} = \mathbb{R}^d$ with the following two properties:

- (a) for any compact $A \subset \mathcal{X}$, N_A has the Poisson distribution with parameter $|A|$ (by which we mean the Lebesgue measure of A), and
- (b) for all $k \in \mathbb{N}$ and pairwise disjoint compact sets $A_1, \dots, A_k$, the random variables $N_{A_1}, \dots, N_{A_k}$ are independent.

For an explicit formula, one can check that if $X_1, X_2, \dots$ are i.i.d. random elements of $\mathcal{X}$ and κ any Poisson random variable, then

$$X = \sum_{n=1}^{\kappa} \delta_{X_n}$$

defines a Poisson process on $\mathcal{X}$. The Poisson process can also be constructed as a limit of Bernoulli processes. Indeed, fix $p \in (0, 1)$ and consider the Bernoulli process X with parameter p on $\mathbb{Z}$. Suppose we shrink our lattice by a factor of p , that is, we replace $\mathbb{Z}$ with $p\mathbb{Z}$. Then the number of lattice points falling in $[0, 1]$, for instance, is roughly $m := 1 + 1/p$, and since each one appears as a point in our process independently of the others with probability p , it follows that

$$\mathbb{P}(N_{[0,1]} = n) = \binom{m}{n} p^n (1-p)^{m-n}.$$

When p is small, m is large and Stirling's formula implies that the right-hand side is approximately

$$\frac{1}{n!} \frac{(m/e)^m}{[(m-n)/e]^{m-n}} p^n (1-p)^{m-n} = \frac{e^{-n}}{n!} \frac{m^m}{(m-n)^{m-n}} p^n (1-p)^{m-n} \approx \frac{e^{-1}}{n!}.$$

In other words, we see that as $p \rightarrow 0$, the Bernoulli process on the compressed lattice $p\mathbb{Z}$ approximates the Poisson process on $\mathbb{R}$. There is a sense in which this can be made precise.

2.2 Correlation Functions

Correlation functions play an analogous role for point processes as do density functions for random variables. More specifically, our running assumption that $\mathcal{X}$ is a “nice” space suggests that there should be some “natural” reference measure μ on it (for instance, if one is thinking of $\mathcal{X}$ as $\mathbb{R}^d$ then μ would be the Lebesgue measure). The correlation functions of a point process X in some sense act as μ -densities for X .

Definition 2.4. Given a point process X on $\mathcal{X}$, its n th correlation measure is the measure ρ_n on $\mathcal{X}^n$ defined as follows by its action on compactly supported Borel-measurable functions $f: \mathcal{X}^n \rightarrow \mathbb{R}$:

$$\int_{\mathcal{X}^n} f(x_1, \dots, x_n) d\rho_n(x_1, \dots, x_n) = \mathbb{E} \left[\sum_{x_1, \dots, x_n \in X} f(x_1, \dots, x_n) \right].$$

Here, summation extends over all n -tuples of distinct points in the process X .

Although not clear from the definition, the sequence $(\rho_n)_{n=1}^\infty$ of all correlation measures determines the process X uniquely; see [here] for a detailed discussion of this. To gain some intuition for the definition, observe that taking $f = \mathbb{I}_A$ to be the indicator of a Borel set A yields

$$\rho_1(A) = \int_{\mathcal{X}} \mathbb{I}_A d\rho_1(x) = \mathbb{E} \left[\sum_{x \in X} \mathbb{I}(x \in A) \right] = \mathbb{E}|X \cap A|,$$

that is, $\rho_1(A)$ measures the expected number of points in A which belong to our process. A similar calculation shows that an analogous interpretation holds for ρ_n with $n > 1$. In particular, for any collection of Borel sets $A_1, \dots, A_n$, we have

$$\rho_n(A_1 \times \dots \times A_n) = \mathbb{E} \left[\prod_{i=1}^n |X \cap A_i| \right] = \mathbb{E}|X^n \cap (A_1 \times \dots \times A_n)|,$$

that is, $\rho_n(A_1 \times \dots \times A_n)$ measures the expected number of points in $A_1 \times \dots \times A_n$ whose components belong to our process.

When our space $\mathcal{X}$ does not have a natural reference measure μ associated with it, the correlation measures are the best we can do in the line of finding a “density function” for the process X . On the other hand, when such a μ does exist, we can improve upon the correlation measures by replacing them with their μ -densities: the *correlation functions*.

Definition 2.5. Given a point process X and a reference measure μ on the same space $\mathcal{X}$, we define the n th correlation function of X , also denoted ρ_n , to be the Radon-Nikodym derivative of the n th correlation measure with respect to μ^n . In other words, ρ_n is the μ^n -a.e. unique function $\mathcal{X}^n \rightarrow \mathbb{R}$ with the property that

$$\rho_n(A) = \int_A \rho_n(x_1, \dots, x_n) d\mu^n(x_1, \dots, x_n)$$

for all Borel sets $A \subset \mathcal{X}^n$.

To gain an understanding of what the correlation functions look like, consider the discrete case $\mathcal{X} = \mathbb{Z}$. Here, the natural reference measure μ is the counting measure and we have that

$$\begin{aligned} \rho_n(x_1, \dots, x_n) &= \int_{\mathcal{X}^n} \mathbb{I}_{\{x_1, \dots, x_n\}} \rho_n(y_1, \dots, y_n) d\mu^n(y_1, \dots, y_n) \\ &= \mathbb{E} \left[\sum_{y_1, \dots, y_n \in X} \mathbb{I}_{\{x_1, \dots, x_n\}}(y_1, \dots, y_n) \right] = \mathbb{P}(x_1, \dots, x_n \in X) \end{aligned}$$

In other words, $\rho_n(x_1, \dots, x_n)$ measures the probability that we see each of the points $x_1, \dots, x_n$ in our process. Analogously, when $\mathcal{X} = \mathbb{R}$ and μ is the Lebesgue measure, one can show that $\rho_n(x_1, \dots, x_n)$ measures the probability that we see a point in each infinitesimal interval $(x_i, x_i + dx_i)$.

In the case when our point process X is an N -point process (recall that this means $|X| = N$ almost surely), we can identify X with N random elements $X_1, \dots, X_N$ of $\mathcal{X}$. If $\mathbb{P}$ is let to denote the joint probability distribution of these N variables, it is natural to assume $\mathbb{P}$ to be symmetric in its arguments (after all, X only cares about *which* points it contains). We can then compute the correlation measures explicitly. Of course, for $n > N$ they vanish identically (if this isn't clear refer back to the definition; the

sum on the right-hand side is necessarily empty in this case). For $n \leq N$, on other hand, we have

$$\begin{aligned} \mathbb{E} \left[\sum_{x_1, \dots, x_n \in X} f(x_1, \dots, x_n) \right] &= \mathbb{E} \left[\sum_{i_1, \dots, i_n \text{ distinct}} f(X_{i_1}, \dots, X_{i_n}) \right] \\ &= \frac{N!}{(N-n)!} \mathbb{E} f(X_1, \dots, X_n) \\ &= \frac{N!}{(N-n)!} \int_{X^N} f(x_1, \dots, x_n) d\mathbb{P}(x_1, \dots, x_N). \end{aligned}$$

Letting $\pi: X^N \rightarrow X^n$ denote the projection map $(x_1, \dots, x_N) \mapsto (x_1, \dots, x_n)$, we can re-write the last expression as

$$\frac{N!}{(N-n)!} \int_{X^N} f(\pi(x_1, \dots, x_N)) d\mathbb{P}(x_1, \dots, x_N) = \frac{N!}{(N-n)!} \int_{X^n} f(x_1, \dots, x_n) d\pi_* \mathbb{P}(x_1, \dots, x_n)$$

and by definition it follows that $\rho_n = \frac{N!}{(N-n)!} \pi_* \mathbb{P}$.

2.3 Determinantal Point Processes

Definition 2.6. A point process on X is said to be *determinantal* if there is a function K on $X \times X$ such that the correlation functions are given by

$$\rho_n(x_1, \dots, x_n) = \det[K(x_i, x_j)]_{i,j=1}^n,$$

that is,

$$\rho_1(x) = K(x, x), \quad \rho_2(x_1, x_2) = \det \begin{bmatrix} K(x_1, x_1) & K(x_1, x_2) \\ K(x_2, x_1) & K(x_2, x_2) \end{bmatrix}, \quad \text{etc.}$$

In this case, the function K is referred to as the *correlation kernel*.

The correlation kernel, if it exists, is non-unique. This is because conjugation $f(y)^{-1}K(x, y)f(x)$ of K by a non-vanishing function $f: X \rightarrow \mathbb{C}$ leaves the determinant unchanged and hence yields a new kernel with the same correlation functions.

Example 2.3. Both Bernoulli and Poisson process are determinantal. To see this in the case of the former, observe that if X is the Bernoulli process with parameter $p \in (0, 1)$ on $X = \mathbb{Z}^d$ then we have

$$\rho_n(x_1, \dots, x_n) = \mathbb{P}(x_1, \dots, x_n \in X) = p^n = \det(pI_{n \times n})$$

so we see that X is determinantal with correlation kernel $K(x_i, x_j) = p\delta_{i=j}$. In the case of the latter, the defining properties of the Poisson process imply (roughly) that

$$\rho_n(x_1, \dots, x_n) = \lim_{\epsilon \rightarrow 0} \mathbb{P}(\text{there is a point in each } [x_i, x_i + \epsilon]) = \lim_{\epsilon \rightarrow 0} \prod_{i=1}^n \epsilon \cdot e^{-\epsilon} = \lim_{\epsilon \rightarrow 0} e^{-n\epsilon + n \log \epsilon} = 1,$$

so we see that X is determinantal with correlation kernel $K(x_i, x_j) = \delta_{i=j}$.

The following class of objects are a fruitful source of determinantal point processes.

Definition 2.7. Given a reference measure μ on X , an N -point *biorthogonal ensemble* on X is an N -point process $\mathbb{P}$ on X satisfying

$$d\mathbb{P} = \frac{1}{Z} \cdot \det[\varphi_i(x_j)]_{i,j=1}^N \cdot \det[\psi_i(x_j)]_{i,j=1}^N d\mu$$

for some choice of functions $\varphi_1, \psi_1, \dots, \varphi_N, \psi_N$ with $|\int_X \varphi_i(x) \psi_j(x) d\mu(x)| < \infty$ for all i and j , and normalizing constant $Z > 0$.

Theorem 2.1. If $\mathbb{P}$ is an N -point biorthogonal ensemble, then $\mathbb{P}$ is a determinantal point process with correlation kernel

$$K(x, y) = \sum_{i,j=1}^N \varphi_i(x) \psi_j(y) G_{ij}^{-\top}, \quad \text{where } G_{ij} = \langle \varphi_i, \psi_j \rangle_{L^2(X, \mu)}.$$

Proof. We first observe that since

$$\begin{aligned}
& \int_{\mathcal{X}^n} \det[\varphi_i(x_j)]_{i,j=1}^N \det[\psi_i(x_j)]_{i,j=1}^N d\mu(x_1) \cdots d\mu(x_N) \\
&= \int_{\mathcal{X}^n} \sum_{\sigma, \tau \in S_N} \operatorname{sgn}(\sigma\tau) \prod_{i=1}^N \varphi_{\sigma(i)}(x_i) \psi_{\tau(i)}(x_i) d\mu(x_1) \cdots d\mu(x_N) \\
&= \sum_{\sigma, \tau \in S_N} \operatorname{sgn}(\sigma\tau) \prod_{i=1}^N G_{\sigma(i)\tau(i)} \\
&= \sum_{\tau \in S_N} \operatorname{sgn}(\tau) \sum_{\sigma \in S_N} \operatorname{sgn}(\sigma) \prod_{i=1}^N G_{\sigma(i)\tau(i)} = \sum_{\tau \in S_N} \det G = N! \det G,
\end{aligned}$$

the normalization constant $Z = N! \det G$ and hence G is invertible. In particular, the given definition of K makes sense. We must now show that

$$\rho_n(x_1, \dots, x_n) = \det \left[K(x_i, x_j) \right]_{i,j=1}^n.$$

In illustration of this, we address two cases. Suppose first that $n > N$. Then the left-hand side vanishes because $\mathbb{P}_N$ is an N -point process. The right-hand side similarly vanishes because the matrix $[K(x_i, x_j)]_{i,j=1}^n$ can be decomposed into a product of rank at most N matrices. Therefore, equality holds and it remains only to address the case $n \leq N$. In this case, the calculation at the end of [Section 2.2](#) implies that

$$\rho_n(x_1, \dots, x_n) = \frac{N!}{(N-n)!Z} \int_{\mathcal{X}^{N-n}} \det[\varphi_i(x_j)]_{i,j=1}^N \det[\psi_i(x_j)]_{i,j=1}^N d\mu(x_{n+1}, \dots, x_N).$$

If we now choose $N \times N$ matrices A and B such that $AGB^\top = I$ and define

$$\Phi_i := \sum_{j=1}^N A_{ij} \phi_j = A_{i,-} \cdot \begin{bmatrix} \phi_1 \\ \vdots \\ \phi_N \end{bmatrix} \quad \text{and} \quad \Psi_i := \sum_{j=1}^N B_{ij} \psi_j = B_{i,-} \cdot \begin{bmatrix} \psi_1 \\ \vdots \\ \psi_N \end{bmatrix},$$

then we have that $\langle \Phi_i, \Psi_j \rangle_{L^2(\mathcal{X}, \mu)} = (AGB^\top)_{ij} = \delta_{ij}$ and also that

$$\det[\Phi_i(x_j)]_{i,j=1}^N = \det A \cdot \det[\varphi_i(x_j)]_{i,j=1}^N \quad \text{and} \quad \det[\Psi_i(x_j)]_{i,j=1}^N = \det B \cdot \det[\psi_i(x_j)]_{i,j=1}^N.$$

The latter implies that we can re-write the formula for the correlation functions as

$$\rho_n(x_1, \dots, x_n) = \frac{1}{(N-n)!} \int_{\mathcal{X}^{N-n}} \det[\Phi_i(x_j)]_{i,j=1}^N \det[\Psi_i(x_j)]_{i,j=1}^N d\mu(x_{n+1}, \dots, x_N).$$

By expanding the determinants, using that $\langle \Phi_i, \Psi_j \rangle_{L^2(\mathcal{X}, \mu)} = \delta_{ij}$, and applying the Cauchy-Binet formula, we obtain that

$$\rho_n(x_1, \dots, x_n) = \det \Phi \Psi^\top$$

for $\Phi = [\Phi_i(x_j)]_{i,j=1}^N$ and $\Psi = [\Psi_i(x_j)]_{i,j=1}^N$. It remains to observe that since $AGB^\top = I$ implies that $G^{-\top} = A^\top B$, expanding according to the definition of the Φ_i and Ψ_i yields that $(\Phi \Psi^\top)_{ij} = K(x_i, x_j)$. $\square$

3 The RS Correspondence and Schur Measures

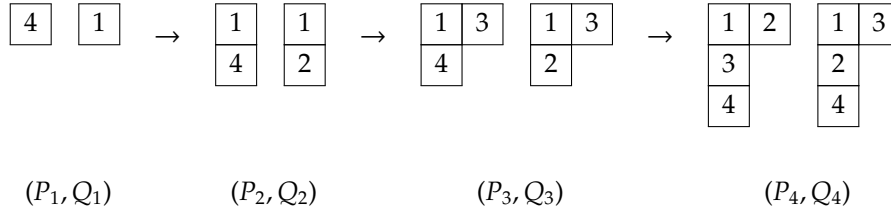
3.1 Motivation

The classical Robinson-Schensted (RS) algorithm takes a permutation $\sigma \in S_n$ of order n and outputs a pair (P, Q) of Young tableaux of the same shape via the following steps:

0. Begin with a pair (P_0, Q_0) of empty tableaux.

- k. Having obtained a pair of Young tableaux (P_{k-1}, Q_{k-1}) , “insert” the pair $(k, \sigma(k))$ to obtain a new pair of tableaux (P_k, Q_k) , as follows. If $\sigma(k)$ is greater than or equal to all numbers in the first row of P_{k-1} , append a box to the end of the first row in each of P_{k-1} and Q_{k-1} , and set their contents to $\sigma(k)$ and k , respectively. If not, then there is a box in the first row of P_{k-1} whose content strictly exceeds $\sigma(k)$; set the content of the first such box (ordered from left to right) to $\sigma(k)$ and “bump down” the replaced content j by inserting it into the second row of P_{k-1} according to the same rules described thus far, potentially resulting in a cascade of “bump downs.” Eventually, a box is appended either to the end of a row in P_{k-1} or below the last row of P_{k-1} (in which case it creates a new row), and the content of this box is the last “bumped” value. In the same location as this box, add a box to Q_{k-1} and set its content to k .

Example 3.1. Applying the above algorithm to the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix}$ yields



Theorem 3.1. The RS algorithm defines a bijection between the set S_n of permutations of order n and the set of pairs (P, Q) of Young tableaux of the same shape λ , where λ ranges over all partitions of n . In particular,

$$\sum_{\lambda \vdash n} (\# \text{ of Young tableaux of shape } \lambda)^2 = n!$$

Moreover, under this correspondence we have the identity

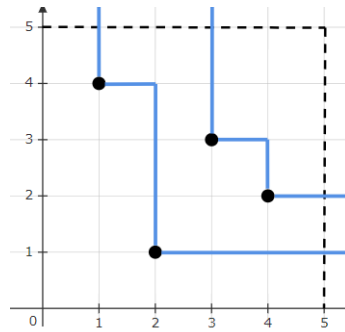
$$\text{length of longest increasing subsequence of } \sigma = \text{length of first row of } P = \text{length of first row of } Q.$$

Proof. See §1.7 of [Rom15]. □

The significance of the RS correspondence stems from its utility in understanding the asymptotic behavior of the longest increasing subsequence of a random permutation, which is of interest due to its relation to growth models in the KPZ universality class. See §1.6-1.20 of [Rom15] as well as [Zyg22] for detailed discussions of this. In this section, we outline an alternative construction of the RS correspondence due to Viennot.

3.2 Viennot’s Geometric Construction of the RS Correspondence

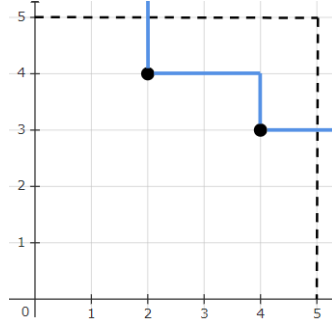
Fix a permutation $\sigma \in S_n$ and consider the realization $S_\sigma = \{(i, \sigma(i)) : 1 \leq i \leq n\}$ of σ as a collection of points in the upper-right quadrant of the plane. For each such point, draw rays emanating from it to the north and to the east, and terminate these rays the first (if any) time they intersect with a ray from another point. Using the permutation from Example 3.1, our picture is the following:



Next, record the values at which the rays hit the dashed lines depicted above and arrange them in increasing order as the first rows of a pair of Young tableaux. In our case, we have

$$\begin{array}{|c|c|} \hline 1 & 3 \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 1 & 2 \\ \hline \end{array}$$

Now replace the original collection of points with the set of points at which the rays intersect and repeat the process thus far, building off the Young tableaux from the previous step. For example, in our case there are two points of intersection among the rays, and if we draw new rays emanating from these points like we did last time, we get the following picture:



By observing where the rays hit the dashed lines, our new pair of Young tableaux becomes

$$\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}$$

The game is to keep doing this until there are no more points of intersection among the rays. In our case, we see that the algorithm will terminate after one more step, yielding the final pair of Young tableaux:

$$\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline 4 & \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline 4 & \\ \hline \end{array}$$

Notice that this is the same pair of Young tableaux we got from the classical RS algorithm! It turns out that this is no coincidence:

Theorem 3.2. *The algorithm described here is equivalent to the RS algorithm; for any permutation $\sigma \in S_n$ they produce the same pair (P, Q) of Young tableaux.*

Proof. See Corollary 3.6.3 of [Sag01]. □

The benefit to Viennot's construction over the classical one is that it renders a lot of the key properties of the algorithm more transparent. For instance, recall that we identify the input permutation σ with a subset of the northeast quadrant of the plane. Clearly, increasing subsequences of σ correspond to upright paths through this subset. On the other hand, the paths formed by the rays move downwards and to the right. It follows that a given increasing subsequence of σ can have at most one point of intersection with each "ray path" and hence

$$\text{length of longest increasing subsequence of } \sigma \leq \text{length of first row of } P = \text{length of first row of } Q.$$

On the other hand, it is clear that by taking the lowest point on each ray path we obtain an increasing subsequence of σ which hits all of them, so in fact the inequality above is an equality. In other words, under Viennot's version of algorithm, this key identity is not only a triviality, but intuitive!

As previously alluded to, the power of the RS algorithm comes from applying it to a *random* permutation. With Viennot's version of the algorithm, we can do something a little more interesting. We've seen that a permutation $\sigma \in S_n$ can be identified with a collection of n points in the northeast quadrant of the plane; namely, the set $\Sigma = \{(i, \sigma(i)) : 1 \leq i \leq n\}$. Of course, this isn't the only set we could've chosen to represent σ by. In fact, all that matters in the context of Viennot's algorithm is the ordering of the horizontal and vertical projections of the points; i.e., we can move the points in Σ around

freely so long as we don't change these orderings. Put differently, we can represent σ by any collection of points in the northeast quadrant of the plane, so long as when we arrange the x and y coordinates of these points in increasing order as $x_1, \dots, x_n$ and $y_1, \dots, y_n$, respectively, the points in our set are precisely $(x_i, y_{\sigma(i)})$.

The upshot of all of this is that, in fact, Viennot's algorithm makes sense (in the sense of being $\text{RS}(\sigma)$ for some permutation σ) when applied to general collections of distinct points in the northeast quadrant of the plane. The natural next step is then to investigate what happens when we replace "general collection of distinct points" with "point process". A particularly interesting case is the following: given a fixed real number $\theta > 0$, consider a Poisson process X in the box B with lower-left corner $(0, 0)$ and upper-right corner (θ, θ) . Let $\lambda(\theta)$ denote the random Young *diagram* (i.e. we forget about the entries of P , as well as the recording tableau Q) produced by applying Viennot's algorithm to this process.

Theorem 3.3. *The distribution of $\lambda(\theta)$ is given by the **Poissonized Plancherel measure** on the set $\mathbb{Y}$ of Young diagrams, that is, for $\mu \in \mathbb{Y}$ we have*

$$\mathbb{P}(\lambda(\theta) = \mu) = e^{-\theta^2} \left(\frac{\theta^{|\mu|} d_\mu}{|\mu|} \right)^2.$$

Here, $|\mu|$ denotes the length of μ and d_μ the number of Young tableaux of shape μ .

Proof. By definition of a Poisson process, the number $n(\theta)$ of points in the box has Poisson distribution with parameter θ^2 . We can therefore identify the points in our process with a random permutation $\sigma(\theta)$ of order $n(\theta)$ (in the manner discussed the preceding paragraphs). Observe that

$$\mathbb{P}(X \subset A \mid n(\theta) = 1) = \frac{\mathbb{P}(X \subset A \text{ and } n(\theta) = 1)}{\mathbb{P}(n(\theta) = 1)} = \frac{\mathbb{P}(N_A = 1, N_{B \setminus A} = 0)}{\mathbb{P}(N_B = 1)} = \frac{|A|e^{-|A|}e^{-(|B|-|A|)}}{|B|e^{-|B|}} = \frac{|A|}{|B|},$$

that is, conditional on there being a single point in our process, the distribution of this point is uniform on B . By conditioning on the locations of $m - 1$ points in our process, one similarly shows that conditional on $n(\theta) = m$ the distribution of the m points in our process is uniform. Clearly this implies that conditional on $n(\theta) = m$ the distribution of the corresponding permutation $\sigma(\theta)$ is uniform on S_m . Since $|\lambda(\theta)| = n(\theta)$, it follows that

$$\mathbb{P}(\lambda(\theta) = \mu) = \mathbb{P}(\lambda(\theta) = \mu \mid n(\theta) = |\mu|) \cdot \mathbb{P}(n(\theta) = |\mu|) = \frac{d_\mu^2}{|\mu|!} \cdot \frac{\theta^{2|\mu|} e^{-\theta^2}}{|\mu|!}$$

as claimed. $\square$

The Poissonized Plancherel measure is an example of a more general class of measures called *Schur measures*, which we discuss in the next section.

3.3 Schur Measures

Consider the algebra Λ of symmetric functions, defined as the set of all formal power series in $\mathbb{C}[x_1, x_2, \dots]$ which are permutation-invariant and of bounded degree, endowed with the usual addition, scalar multiplication, and product operations. An extremely important class of functions belonging to this algebra are the *Schur polynomials*, defined as follows.

Definition 3.1. The *Schur polynomials* are a family of polynomials $\{s_\lambda\}_{\lambda \in \mathbb{Y}}$ indexed by the set of Young diagrams, defined for a given diagram $\lambda = (\lambda_1, \dots, \lambda_n)$ by

$$s_\lambda(x_1, \dots, x_n) = \frac{\det [x_i^{\lambda_j + n - j}]_{i,j=1}^n}{\det [x_i^{n-j}]_{1 \leq i,j \leq n}}.$$

While the above definition is the original one given by Schur, the Schur polynomials can alternatively be defined combinatorially as follows: for each Young diagram $\lambda = (\lambda_1, \dots, \lambda_n)$, we set

$$s_\lambda(x_1, \dots, x_n) = \sum_T x_1^{\# \text{ of } 1\text{'s in } T} \dots x_n^{\# \text{ of } n\text{'s in } T}$$

where summation extends over all Young tableaux T of shape λ . This definition might seem slightly more natural than the first, but the determinantal expression is preferred as it allows one to express the laws of certain integrable models in terms of determinants, which in turns renders their asymptotic analysis a much easier task.

The reason why Schur polynomials are of interest to us is that they give rise to determinantal point processes. In particular, let us say that an algebra homomorphism $\varphi: \Lambda \rightarrow \mathbb{C}$ is *Schur-positive* if for any Young diagram λ we have that $\varphi(s_\lambda) \geq 0$. Given a pair φ, ψ of Schur-positive homomorphisms, we can associate to them a probability measure $\mathbb{P}_{\varphi, \psi}$ on the set of Young diagrams defined by

$$\mathbb{P}_{\varphi, \psi}(\lambda) = \frac{\varphi(s_\lambda)\psi(s_\lambda)}{N(\varphi, \psi)}, \quad N(\varphi, \psi) = \sum_{\lambda} \varphi(s_\lambda)\psi(s_\lambda)$$

provided the series on the right converges. Because Young diagrams can be identified with point configurations in $\mathbb{Z} + \frac{1}{2}$ via the correspondence

$$\lambda = (\lambda_1, \dots, \lambda_n) \iff X_\lambda := \left\{ \lambda_i - i + \frac{1}{2} \right\}_{i=1}^{\infty}$$

(on the right-hand side, we interpret λ_i as being 0 for $i > n$) it follows that Schur measures define point processes on $\mathbb{Z} + \frac{1}{2}$. In fact, these point processes turn out to be determinantal (though perhaps this is unsurprising given how we defined the Schur polynomials).

Theorem 3.4. *If $\lambda \in \mathbb{Y}$ is distributed according to a Schur measure $\mathbb{P}_{\varphi, \psi}$, then X_λ is a determinantal point process on $\mathbb{Z} + \frac{1}{2}$.*

Proof. We have that

$$\rho_n(x_1, \dots, x_n) = \mathbb{P}(x_1, \dots, x_n \in X) = \sum_{\lambda: x_1, \dots, x_n \in X_\lambda} \mathbb{P}_{\varphi, \psi}(X_\lambda) = \frac{1}{N(\varphi, \psi)} \sum_{\lambda: x_1, \dots, x_n \in X_\lambda} \varphi(s_\lambda)\psi(s_\lambda).$$

Use the definition of the Schur polynomials to now argue that $\mathbb{P}_{\varphi, \psi}$ is in fact a biorthogonal ensemble. The result then follows from [Theorem 2.1](#). In fact, this gives us a correlation kernel for $\mathbb{P}_{\varphi, \psi}$ as well. See [\[BG15\]](#) for details. $\square$

Corollary 3.1. *The Poissonized Plancherel measure with parameter $\theta > 0$ on the set of Young diagrams is a determinantal point process on $\mathbb{Z} + \frac{1}{2}$.*

Proof. By Proposition 2.4 of [\[BG15\]](#), the Schur polynomials form a basis for Λ . Therefore, if we define $\varphi: \Lambda \rightarrow \mathbb{C}$ by setting

$$\varphi(s_\lambda) = \frac{\theta^{|\lambda|}}{|\lambda|!} d_\lambda$$

and extending via the homomorphism property, we get a Schur-positive homomorphism. It is easy to see that the associated Schur measure is the Poissonized Plancherel measure with parameter θ . $\square$

The significance of the above is that we can understand the asymptotic behavior of the Poissonized Plancherel measure by instead understanding the behavior of its correlation kernel as the parameter $\theta \rightarrow \infty$. This can be done via the method of steepest descent, which we carry out in the next section.

4 Asymptotic Behavior of the Poissonized Plancherel Measure

The results of the last section say that the Poissonized Plancherel measure with parameter $\theta > 0$ is a determinantal point process on $\mathbb{Z} + \frac{1}{2}$, and in particular a biorthogonal ensemble. From [Theorem 2.1](#), we thus have an explicit formula for the correlation kernel $K_\theta(i, j)$, which can be re-formulated in terms of contour integration as follows:

$$K_\theta(i, j) = -\frac{1}{4\pi^2} \oint_{|w|=1} \oint_{|v|=2} \exp \left[\theta \left(v - \frac{1}{v} - w + \frac{1}{w} \right) \right] \cdot \frac{\sqrt{vw}}{(v-w)v^{i+1}w^{-j+1}} dv dw;$$

see [BG15] for details regarding the derivation of this formula. In this section, we present a steepest descent analysis of the asymptotic behavior of $K_\theta(i, j)$ as $\theta \rightarrow \infty$ due to Okounkov [Oko03]. We moreover present only the case $i = j$; for the general case, see [BOO99].

Since $K_\theta(i, i)$ measures the “density” of the points in our process, it is natural to make the change of variables $i = u\theta$, whence K_θ becomes

$$K_\theta(u\theta, u\theta) = -\frac{1}{4\pi^2} \oint_{|v|=1} \oint_{|w|=2} \exp[\theta(S(v) - S(w))] \cdot \frac{1}{(v-w)\sqrt{vw}} dv dw$$

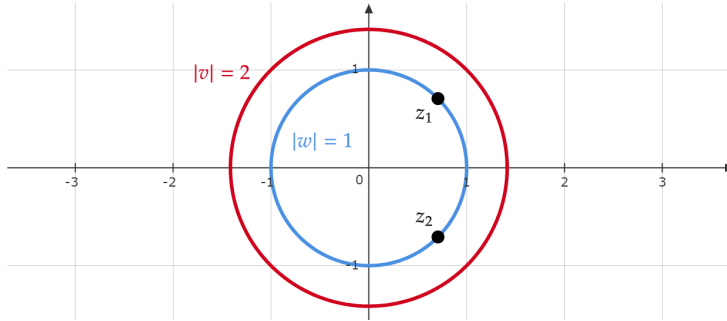
for $S(z) = z - \frac{1}{z} - u \log z$. As is the usual objective in steepest descent analyses, we wish to deform the above contours so that the values of the integrals are essentially determined by what’s happening at the critical points of $S(z)$. Of course, in our case we’ll need to proceed with slightly more care than usual, since the $\log z$ appearing in the definition of $S(z)$ and $\sqrt{vw}$ term render our integrand (which shall henceforth be denoted by $f(v, w)$ to simplify notation) non-holomorphic. With this in mind, we first identify the critical points of $S(z)$. Indeed, if we take the principal branch of $\log$, then

$$S'(z) = 1 + \frac{1}{z^2} - \frac{u}{z}$$

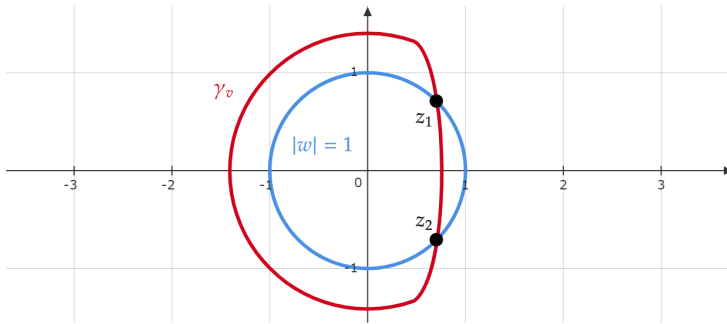
exists outside of $(-\infty, 0]$ and has two roots:

$$z_1 = \frac{u + \sqrt{u^2 - 4}}{2} \quad \text{and} \quad z_2 = \frac{u - \sqrt{u^2 - 4}}{2}$$

When $|u| < 2$, these become complex conjugated roots of unit modulus; hence they are of the form $e^{\pm i\phi}$ for some ϕ . In fact, one easily checks that ϕ is characterized by $2 \cos \phi = u$. Our picture thus looks something like the following:



What we’d like to do now is deform the contour $|v| = 2$ so that it passes through the critical points z_1 and z_2 , that is, we’d like to deform $|v| = 2$ into something like



Of course, we can’t actually do this because (i) the function we’re integrating is not complex analytic everywhere in region of interest, so Cauchy’s theorem doesn’t apply here, and (ii) even if it did,

deforming the contours like this would still be illegal on account the $v - w$ term in the denominator. Despite these issues, however, each of

$$\oint_{|w|=1} \oint_{|v|=2} f(v, w) dv dw \quad \text{and} \quad \oint_{|w|=1} \oint_{\gamma_v} f(v, w) dv dw$$

are still defined, we just can't evaluate nor manipulate them via standard complex analytic techniques. Recall that our goal is to understand the behavior of the former as $\theta \rightarrow \infty$. By steepest descent, we can easily do this for the latter. We'll therefore be able to do the same for the former, provided we can understand how the two *differ*. In other words, what we're interested in is only how the value of the former changes as we make the (illegal) transformation of $|v| = 2$ into γ_v , that is, the value of

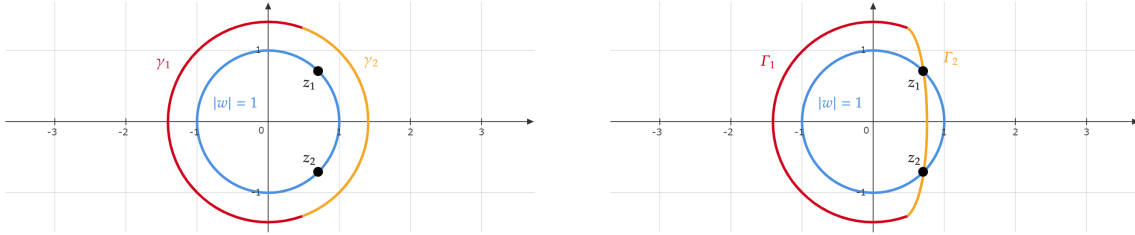
$$\oint_{|w|=1} \left(\oint_{|v|=2} f(v, w) dv - \oint_{\gamma_v} f(v, w) dv \right) dw.$$

What's nice about this is that (roughly) because the contours $|v| = 2$ and γ_v aren't too different, cancellation will occur in the areas where $f(v, w)$ is poorly behaved, allowing us to sidestep all of the aforementioned issues regarding complex analyticity.

To see this concretely, fix $w \in S^1$. We wish to evaluate

$$\oint_{|v|=2} f(v, w) dv - \oint_{\gamma_v} f(v, w) dv.$$

Let us take the principal branch of $\sqrt{\cdot}$, so that $f(v, w)$ is defined and holomorphic outside of the negative reals, with singularities at the origin and w . Let us moreover break up the contours $|v| = 2$ and γ_v into the following unions of disjoint paths:



Since $\gamma_1 = \Gamma_1$, the difference of the two integrals reduces to the integral of $f(v, w)$ over the closed contour formed by γ_2 and Γ_2 . Notice that $f(v, w)$ is holomorphic in the region enclosed by this contour. In particular, if w does not belong to the arc between z_1 and z_2 , then $f(v, w)$ has no singularities in the enclosed region and hence

$$\oint_{|v|=2} f(v, w) dv - \oint_{\gamma_v} f(v, w) dv = 0$$

by the residue theorem. On the other hand, if w does belong to the arc between z_1 and z_2 , then w is the only singularity of $f(v, w)$ in the enclosed region and hence the difference evaluates to $2\pi i \operatorname{Res}(f(v, w), w)$. It follows that

$$\begin{aligned} -\frac{1}{4\pi^2} \oint_{|w|=1} \left(\oint_{|v|=2} f(v, w) dv - \oint_{\gamma_v} f(v, w) dv \right) dw &= \frac{1}{(2\pi i)^2} \oint_{|w|=1} 2\pi i \operatorname{Res}(f(v, w), w) \mathbb{I}(w \in A) dw \\ &= \frac{1}{2\pi i} \int_{e^{-i\phi}}^{e^{i\phi}} \frac{dz}{z} \\ &= \frac{\phi}{\pi} \end{aligned}$$

and hence that

$$K_\theta(u\theta, u\theta) = -\frac{1}{4\pi^2} \oint_{|w|=1} \oint_{\gamma_v} f(v, w) dv dw + \frac{\phi}{\pi}.$$

The hard part is now over. To complete the analysis, we claim that the first term on the right-hand side decays as $\theta \rightarrow \infty$. In particular, we claim that $\Re(S(v) - S(w)) < 0$. Since we have that

$$\exp[\theta(S(v) - S(w))] = \exp[\theta \Re(S(v) - S(w))] \cdot \exp[i\theta \Im(S(v) - S(w))],$$

the second term of which is bounded in θ , this indeed implies the claimed decay.

We first observe that $\Re(S(z))$ vanishes along $|z| = 1$ since for every such z we have that $z = e^{i\alpha}$ for some $\alpha \in [0, 2\pi)$, whence

$$S(z) = e^{i\alpha} - e^{-i\alpha} - u \log e^{i\alpha} = 2i \sin \alpha - i\alpha u \in i\mathbb{R}$$

In addition, one easily computes that for $|z| = 1$ we have

$$\nabla \Re(S(z)) = (2\Re(z) - u)z = (2 \cos \alpha - 2 \cos \phi)z.$$

In other words, when $\alpha < \phi$ (in which case $2 \cos \alpha - 2 \cos \phi > 0$) the gradient vector point outwards, and when $\alpha > \phi$ it points inwards. Clearly this implies that $\Re(S(z)) < 0$ along γ_v , so we have that

$$\Re(S(v) - S(w)) = \Re(S(v)) - \Re(S(w)) = \Re(S(v)) < 0$$

as claimed. In light of the aforementioned decay, we conclude that

$$\lim_{\theta \rightarrow \infty} K_\theta(u\theta, u\theta) = \frac{\phi}{\pi}.$$

This can actually be realized as an instance of the Vershik–Kerov–Logan–Shepp limit shape theorem; see [BG15] for further discussion.

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